The Journal of Open Source Software

3D ICE: An Interactive Browser-Based Cryosphere Explorer for Antarctica and Greenland

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DOI: 10.xxxxxx/draft

Software

- Review
- Repository
- Archive

Editor:

Submitted: 28 September 2026
Published: unpublished

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Summary

3D ICE is an open-source platform for exploring research-grade representations of the Antarctic and Greenland ice sheets in a web browser (Figure 1). It combines terrain, ice velocity, ocean circulation, ice-shelf basal melting, basal friction, subglacial hydrology, and drainage basins in layered three-dimensional scenes. Users can rotate and zoom each ice sheet, switch datasets and resolutions, search polar places, show the ice-free bed after isostatic rebound, play multi-model projections of Antarctic ice to 2300, and follow links to the source data. A research edition offers every dataset and layer; a public edition for classrooms and outreach pairs a subset with a guided tour and plain-language explanations.

An offline Python pipeline converts heterogeneous NetCDF, HDF5, GeoTIFF, and vector products into compact binary packages with provenance metadata, which a static JavaScript application built with Three.js (Cabello & three.js contributors, 2024) and WebGL renders without plugins or server-side computation, on desktop and mobile devices.

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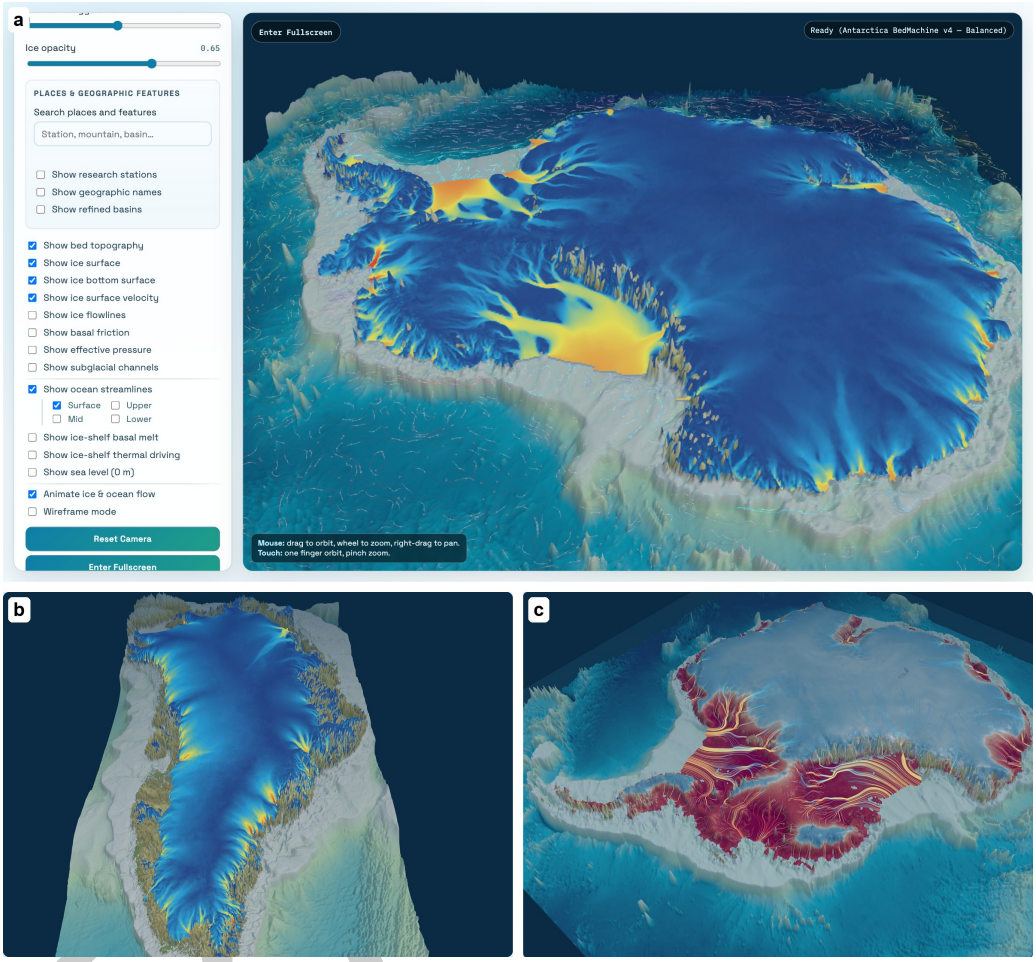


Figure 1: The 3D ICE explorer. (a) The interface, with Antarctic ice-surface speed (Mouginot et al., 2019a, 2019b) and WAOM2 ocean streamlines seeded in the top 3–18 % of the local water depth (Boeira Dias et al., 2023; Richter et al., 2022) over BedMachine Antarctica v4 (Morlighem et al., 2020; Morlighem, 2026). (b) Greenland ice-surface speed from the ITS_LIVE V02.1 summary mosaic (2014–2024), generated using auto-RIFT (Gardner et al., 2018) and provided by the NASA MEaSUREs ITS_LIVE project (Gardner et al., 2025), over BedMachine Greenland v6 (Morlighem et al., 2017, 2025). (c) Antarctic ice in 2300 under high emissions (SSP5-8.5), the mean of eight ISMIP6 models (Seroussi, Pelle, Lipscomb, et al., 2024), coloured by thickness change since 2015 (red: thinning), with flowlines coloured by speed, at the public edition's high-emission tour stop. Vertical scales are exaggerated.

Statement of need

Modern cryosphere science generates large, gridded datasets in specialized formats and polar stereographic coordinate systems, and ice-sheet projections add terabyte-scale multi-model archives. Inspecting several products together commonly requires a desktop GIS, a numerical computing environment, or purpose-built scripts. These workflows suit quantitative analysis but require installation, data transfer, reprojection, and preprocessing before a user can see an ice sheet in context.

The barrier falls hardest on researchers in adjacent disciplines, educators who want to teach with real polar data, and scientists preparing interactive research communication, who need to see quickly how ice geometry, flow, ocean forcing, and basal processes relate, with a path back to the source data. 3D ICE provides a curated, zero-install view, not a replacement for GIS or numerical analysis; its research purpose is to make heterogeneous cryosphere products easier

to compare, explain, and select for analysis.

State of the field

Quantarctica (Matsuoka et al., 2021) and its Greenland counterpart QGreenland (Moon et al., 2023) are comprehensive data packages and mapping environments for desktop QGIS. They suit local geospatial analysis and span more disciplines than 3D ICE, which trades general GIS operations for an immediately shareable, curated 3D view with one interaction model for both ice sheets.

NASA Worldview offers web access to more than a thousand global satellite-imagery layers, with polar views, temporal comparison, animation, and data download (NASA Earth Science Data and Information System, 2026), but focuses on two-dimensional, often near-real-time imagery. 3D ICE instead combines terrain with subsurface and model-derived fields such as basal friction, hydrology, basal melt, depth-dependent ocean circulation, and projected ice geometry.

General 3D libraries such as CesiumJS (Cesium GS, Inc. and CesiumJS contributors, 2026), with its high-precision WGS84 globe and scalable data formats, and Three.js (Cabello & three.js contributors, 2024), the lower-level graphics that 3D ICE uses, supply no cryosphere data model, polar-grid conversion, provenance records, layer semantics, or data curation. Contributing these product-specific transformations to a general rendering engine would not address the research need, so 3D ICE reuses established graphics infrastructure and concentrates its scholarly contribution in the reproducible polar-data pipeline, the data contract, and domain-specific interaction design.

Software design

3D ICE has two stages joined by a specified data contract. The offline stage reads authoritative source products, reconciles their coordinate conventions, and resamples fields onto terrain-aligned polar grids with NumPy (Harris et al., 2020). Floating-point grids are quantized to signed 16-bit arrays for browser delivery, a controlled trade of precision for transfer size: a paired `.meta.json` file stores scale, offset, fill value, units, grid geometry, statistics, source citation, and processing provenance, so decoding is deterministic and the loss of precision inspectable. One decoder module reads this metadata in the page, in the geometry worker, and in a test that checks every committed package against the statistics recorded before quantization, to within half a quantization step.

Expensive transformations run before deployment, so the application needs only static hosting; the trade-off is that updating a source product means regenerating its package. Antarctic ocean streamlines, for example, are integrated through multi-depth WAOM2 velocity fields with seeds balanced by region and depth (Boeira Dias et al., 2023; Richter et al., 2022). The projection layer plays the equal-weight mean of eight ISMIP6 Antarctica 2300 models, one per modelling group, under three UKESM1-0-LL scenarios: low emissions, high emissions, and high emissions with ice-shelf collapse (Jourdain et al., 2020; Nowicki et al., 2020; Nowicki & ISMIP6 Team, 2024; Payne et al., 2021; Seroussi, Pelle, Lipscomb, et al., 2024). Scripts beside the 8 TB archive on a supercomputer regrid each model conservatively. A packer applies the models' mean thickness change since 2015 to today's BedMachine ice at 5-year keyframes, scaling thinning to BedMachine's thickness so that ice shelves do not break into spurious holes, and stores their mean speed change. The browser interpolates between keyframes, rebuilds the ice by flotation on the static bed, and moves the flowlines, coloured by today's speed plus that change, onto the projected surface. The averaged geometry implies more sea-level rise than the mean of the models' published contributions (1.93 against 1.46 m by 2300 under high emissions), partly because flotation is non-linear where the models disagree about which basins

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collapse, so the explorer reports the published mean and range (Seroussi, Pelle, & ISMIP6 Team, 2024).

Web Workers build overlay geometry off the main thread, and Balanced and HD packages trade resolution against memory. Logic that needs neither page nor scene, such as package decoding, place search, the rebound solver, and projection playback, lives in modules that run unchanged under Node's test runner. Both editions, in English and Chinese, are thin shells around one runtime: an edition profile removes restricted datasets and layers, with their package URLs, before anything is fetched, and pins the controls a page omits to fixed states, so the public pages can neither request nor show a research-only layer.

The isostatic-rebound layer shows which parts of today's sub-sea-level bed would emerge once glacial isostatic adjustment is complete. By default it displays the published response of Paxman et al. (2022, 2026): elastic-plate flexure with laterally variable elastic thickness (Steffen et al., 2018; Swain & Kirby, 2021), remaining post-LGM disequilibrium, and water loading under a sea surface raised by both ice sheets' meltwater. For comparison it solves an elastic-lithosphere, relaxed-asthenosphere steady state (Le Meur & Huybrechts, 1996; Lingle & Clark, 1985) spectrally (Bueler et al., 2007) and, as an upper bound on peak uplift, local Airy isostasy (Turcotte & Schubert, 2002). Sea-level-equivalent figures follow Gregory et al. (2019). The interface states each response's assumptions, including that the present bed is not in balance with the present load (Whitehouse et al., 2019) and that response times are orders of magnitude shorter over the low-viscosity mantle beneath parts of West Antarctica (Barletta et al., 2018).

Unit tests cover quantization, coordinate sampling, metadata statistics, search, projection playback, edition profiles, the tour and its copy in both languages, and the rebound solver against the analytic Kelvin-function point-load solution (Brotchie & Silvester, 1969); integration tests regenerate derived packages and compare them with the committed files; and Playwright tests load both editions, check that the public edition requests no research-only package, and walk its tour stop by stop. All run in continuous integration and locally.

Research impact statement

Y. Wang uses 3D ICE to compare candidate bed, velocity, and basal-friction products before configuring ice-flow and subglacial-hydrology simulations, and has used its views in a research funding proposal. It has been demonstrated in talks at the Antarctic Research Centre, Victoria University of Wellington (February 2026), the ISMIP7 Workshop in Copenhagen (March 2026, presented by C. Zhao), the Asia Early Career Polar Forum in Zhuhai (June 2026), and the School of Oceanography, Shanghai Jiao Tong University (July 2026).

Y. Lin has used 3D ICE in teaching GE1301 Climate Change and Extreme Weather at the City University of Hong Kong, to show students the geography, ice flow, and surrounding ocean of Antarctica and Greenland, and is scheduled to demonstrate it to the public at InnoCarnival 2026 (Hong Kong Science Park, October–November 2026). F. Boeira Dias, who provided the WAOM2 output that 3D ICE renders, has described using the explorer to visualise ocean circulation that is difficult to interpret in standard two-dimensional plots (Australian Antarctic Program Partnership, 2026; Spatial Source, 2026). From its launch on 22 March to 27 September 2026, the public site recorded 578 active users and about 1,200 page views.

3D ICE integrates representative products for bed geometry (Morlighem et al., 2017, 2020, 2025; Morlighem, 2026; Palmer et al., 2025; Pritchard et al., 2024, 2025), ice velocity (Gardner et al., 2018, 2025; Mouginot et al., 2019a, 2019b), ocean circulation (Boeira Dias et al., 2023; Copernicus Marine Service, 2025; Richter et al., 2022), ice-shelf basal melting (Galton-Fenzi et al., 2025; Galton-Fenzi, 2025), basal friction (Jager, Gladstone, Zwinger, Wang, et al., 2026; Jager, Gladstone, Zwinger, Gillet-Chaulet, et al., 2026; Jager, 2026a, 2026b), subglacial hydrology (Ehrenfeucht, 2024; Ehrenfeucht et al., 2025; Werder et al., 2013), drainage boundaries (Mouginot et al., 2017; Mouginot & Rignot, 2019; Rignot et al.,

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2011), and polar places (Council of Managers of National Antarctic Programs, 2025; Scientific Committee on Antarctic Research, 2026).

The repository supplies a specified data contract, a documented preparation pipeline, tagged releases with a changelog and a compatibility bundle (Wang & Lin, 2026), contribution and support pathways, and worked examples that run from a clean clone: one reproduces the explorer's Antarctic rebound figures, including 2.94 million km^2 of today's sub-sea-level bed emerging above the ice-free sea surface, and another regenerates six derived data packages whose payloads match the committed files byte for byte.

AI usage disclosure

Generative AI was used extensively in developing 3D ICE. The first prototype of the explorer, and much of its subsequent application code, was generated with the OpenAI Codex coding agent (GPT-5-family models, February to August 2026) from Y. Wang's natural-language specifications, data, and review, as the project's public coverage describes (Australian Antarctic Program Partnership, 2026). Anthropic's Claude Code (Claude Opus 4.6 in March and April 2026; Claude Opus 5 and Claude Opus 5.5 in September and October 2026) was used for later work: parts of the interface, including the public edition and the home pages; test infrastructure; the isostatic-rebound and projection layers and their data pipelines; the refactoring of the runtime into shared, tested modules; the repository documentation; reference checking; and the drafting and copy-editing of this paper. Y. Wang selected the datasets and scientific methods and set the architecture. Every AI-generated step was checked by hand before it was kept: Y. Wang read each change, ran the explorer to confirm its behaviour, validated each scientific layer against its source product, and checked every reference against its source. The automated checks described above back this review. The authors accept responsibility for the accuracy, originality, licensing, and integrity of the submitted work.

Author contributions

Y. Wang conceived, designed, and developed 3D ICE and wrote the paper. Y. Lin helped debug and evaluate the explorer, advised on the isostatic-rebound and sea-level calculations and on the interface design for public audiences, and joined discussions of its development.

Acknowledgements

3D ICE has received no dedicated funding. It was begun at the Institute for Marine and Antarctic Studies, University of Tasmania, while Y. Wang held an Australian Antarctic Program Partnership top-up scholarship and a Tasmanian Graduate Research Scholarship, and continued at the Climate Systems Engineering initiative, University of Chicago. The authors gratefully acknowledge the providers of the data that make 3D ICE possible: the National Snow and Ice Data Center, the UK Polar Data Centre, ITS_LIVE, the Copernicus Marine Service, the Australian Antarctic Division, the NSF Arctic Data Center, SCAR, COMNAP, and the authors of the QRF, WAOM2, GlaDS, and basal-friction products. They also acknowledge the World Climate Research Programme for coordinating CMIP6, the climate and ice-sheet modelling groups, the Earth System Grid Federation, and ISMIP6 and Ghub for the projections. This study has been conducted using E.U. Copernicus Marine Service Information; https://doi.org/10.48670/moi-00001.

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